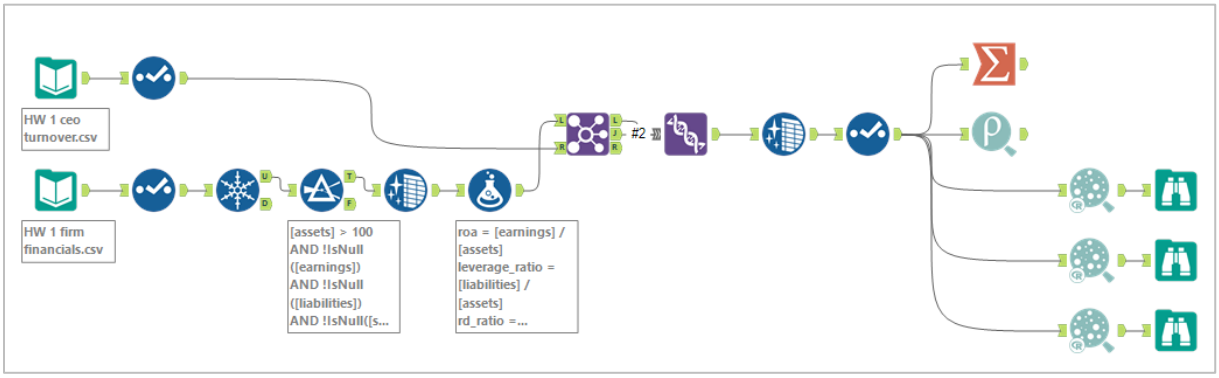
Homework 1 - Alteryx
Accounting Analytics 5559
Henry Laudano
9/22/2026

Exploratory Analysis



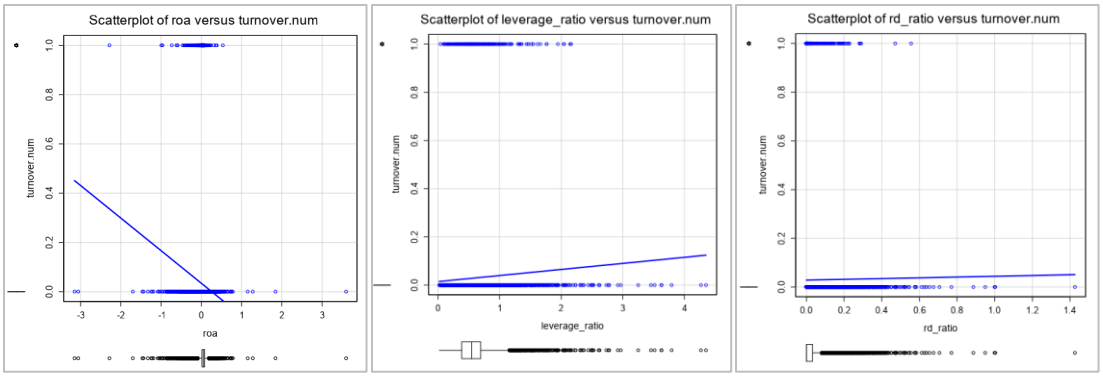
1) Averages & Correlations

Record	Avg_roa	Avg_leverage_ratio	Avg_rd_ratio	turnover
1	0.041903	0.557951	0.030162	0
2	-0.032145	0.627394	0.032404	1

Record	FieldName	roa	leverage_ratio	rd_ratio	turnover
1	roa	1	-0.160063	-0.203004	-0.098961
2	leverage_ratio	-0.160063	1	-0.14448	0.041794
3	rd_ratio	-0.203004	-0.14448	1	0.005926
4	turnover	-0.098961	0.041794	0.005926	1

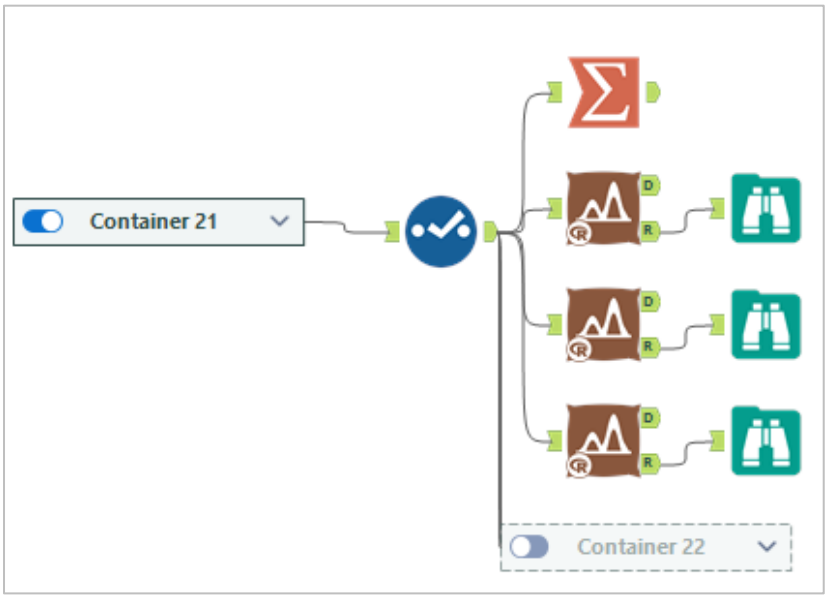
The averages suggest that firms with CEO turnover have, on average, a lower return on assets (−0.032 vs. 0.042), a higher leverage ratio (0.627 vs. 0.558), and a relatively similar R&D to assets ratio (0.032 vs. 0.030). This is supported by the correlations with turnover, which are slightly negative (r = −0.099), slightly positive (r = 0.042), and less than 0.01 (r = 0.006) for return on assets, leverage, and R&D, respectively.

2) Scatterplots



Such correlations are not visualized very well in the scatterplots, because they are binary – these relationships can be investigated further using statistical analysis.

Statistical Analysis



3) Test of Means

Record	Report
1	**Welch's Two Sample t-test(s) of roa by turnover**

Test	t-Statistic	Degrees of Freedom	p-Value
1 vs 0	7.79022	410.93	5.5188e-14

Record	Report
1	**Welch's Two Sample t-test(s) of leverage_ratio by turnover**

Test	t-Statistic	Degrees of Freedom	p-Value
1 vs 0	-4.31747	419.03	1.9719e-05

Record	Report
1	**Welch's Two Sample t-test(s) of rd_ratio by turnover**

Test	t-Statistic	Degrees of Freedom	p-Value
1 vs 0	-0.685123	424.01	0.49364

The two-sample t-tests confirm the trend shown from the averages, where both return on assets ($t = 7.79$, $p < .001$) and leverage ratio ($t = -4.32$, $p < .001$) have p values $< .01$, showing a lower average return on assets when there is turnover, and a higher average leverage ratio. The R&D ratio difference in the two means was insignificant ($t = -0.69$, $p = .494$).

Predictive Analysis

4) Logistic Regression

Report

Report for Logistic Regression Model logistic_regression_turnover

Basic Summary

Call:

glm(formula = turnover ~ roa + leverage_ratio + rd_ratio + ln_assets, family = binomial("logit"), data = the.data)

Deviance Residuals:

Min	1Q	Median	3Q	Max
-3.156	-0.246	-0.230	-0.214	3.078

Coefficients:

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	-3.17076	0.27507	-11.527	< 2.2e-16	***
roa	-2.32208	0.30362	-7.648	2.04e-14	***
leverage_ratio	0.48729	0.16062	3.034	0.00242	**
rd_ratio	-1.11610	0.72009	-1.550	0.12115	
ln_assets	-0.07197	0.03484	-2.066	0.03884	*

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial taken to be 1)

Null deviance: 3644.7 on 13963 degrees of freedom
Residual deviance: 3553.4 on 13959 degrees of freedom
McFadden R-Squared: 0.02506, Akaike Information Criterion 3563

Number of Fisher Scoring iterations: 6

Type II Analysis of Deviance Tests

The directional estimates of the coefficients and their significance line up with the exploratory and statistical analysis done earlier, with the significant relationships of return on assets having a negative relationship with turnover ($\beta = -2.32$, $z = -7.65$, $p < .001$) and leverage ratio having a positive one ($\beta = 0.49$, $z = 3.03$, $p = .002$). R&D ratio was not a significant predictor ($\beta = -1.12$, $z = -1.55$, $p = .121$). The new predictor, ln(assets), was a significant negative predictor ($\beta = -0.07$, $z = -2.07$, $p = .039$), added as another predictor suggesting that larger firms have less CEO turnover, holding the other variables constant.

Intuitively, this makes sense – struggling or poorly returning firms would likely fire their CEOs, and return on assets is a good measure of this performance. These struggling firms might be more in financial distress, reflecting the leverage ratio relationship. Larger firms might be more successful with higher-quality CEOs, reflecting a lower need for them to be fired. Finally, R&D ratio is likely just an indicator of different business models, suggesting no relationship there.

5) Neural Network Model Comparison

Record	Model	Accuracy	Accuracy_0	Accuracy_1	F1	AUC
1	logistic_regression_turnover	0.970997	0.999705	0.002488	0.004914	0.667472
2	neural_network_turnover	0.971212	1	0	[Null]	0.696245

Both models have essentially the same high performance (neural network very marginally higher accuracy, 97.12% vs. 97.10%). The logistic regression model has a higher F1 (0.005 vs. null) whereas the neural network has a higher AUC (0.696 vs. 0.667).

6) Prediction vs. Averages

Record	roa	leverage_ratio	rd_ratio	ln_assets	Score_0	Score_1
1	-0.2	0.8	0.05	5	0.866596	0.133404

Record	Avg_turnover	Avg_roa	Avg_leverage_ratio	Avg_rd_ratio	Avg_ln_assets
1	0.028788	0.039771	0.55995	0.030227	7.793258

Since the neural network had both higher accuracy and a higher AUC, it was selected as the better-performing model and used for the prediction. The predicted turnover probability from the neural network was 13.3%, which is higher than the average turnover of 2.9%.

Based on this finding, yes, the CEO should be worried about their job, as their risk of being fired is higher. I would recommend the CEO both work on bringing the company to profitability (raise the ROA) and to de-lever as priorities based on this model, but it is important to note these models are observational and do not indicate causation.